$\pi_u = 13.5\Delta - 6.5$

$u = 1.35$

$\pi_0 = 1 \times 10 - f$

$\pi_d = 7.4\Delta - 0.4$

$d = 0.74$

{ long Δ shares

short 1 call.

$\pi = \Delta \cdot S - C$

$1 \times 10 - f = 6.6586$

$f = 3.34$

$S\uparrow$ $C\downarrow$

long share + long call.

$S\uparrow$	$C\uparrow$
$S\downarrow$	$C\downarrow$

$13.5\Delta - 6.5 = 7.4\Delta - 0.4$

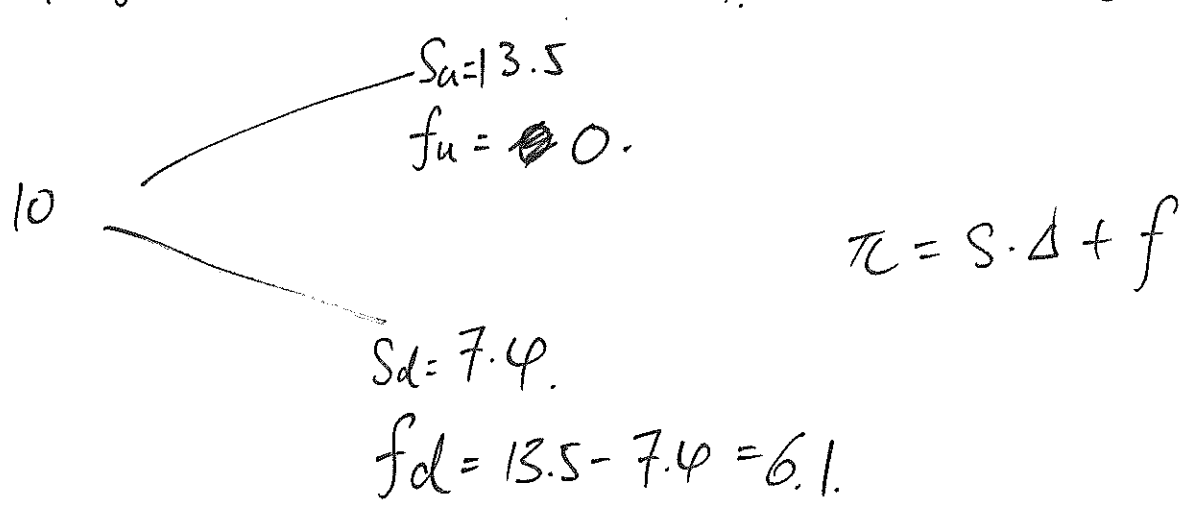
$6.1\Delta = 6.1$

$\Delta = 1$

{ short Δ shares ↓ π = -ve.

Long 1 call. ↑

payoff = max $(K - S_T, 0)$ $K = 13.5$.



long Δ shares. ↑ long put.

long 1 put. ↓ S↑ ↓

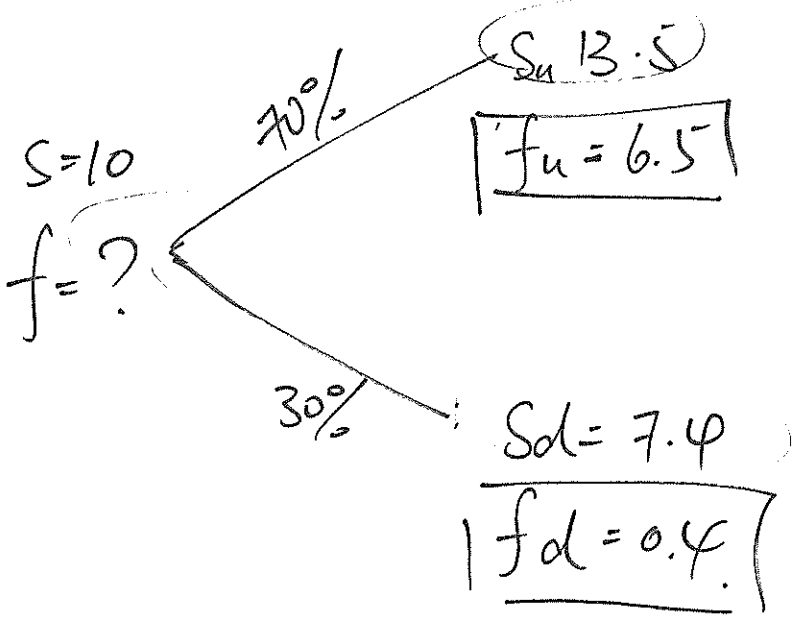
S↑ $\pi_u = \pi_d = 13.5\Delta + 0 = 7.4\Delta + 6.1$

$\Delta = 1$

$\pi = 13.5$

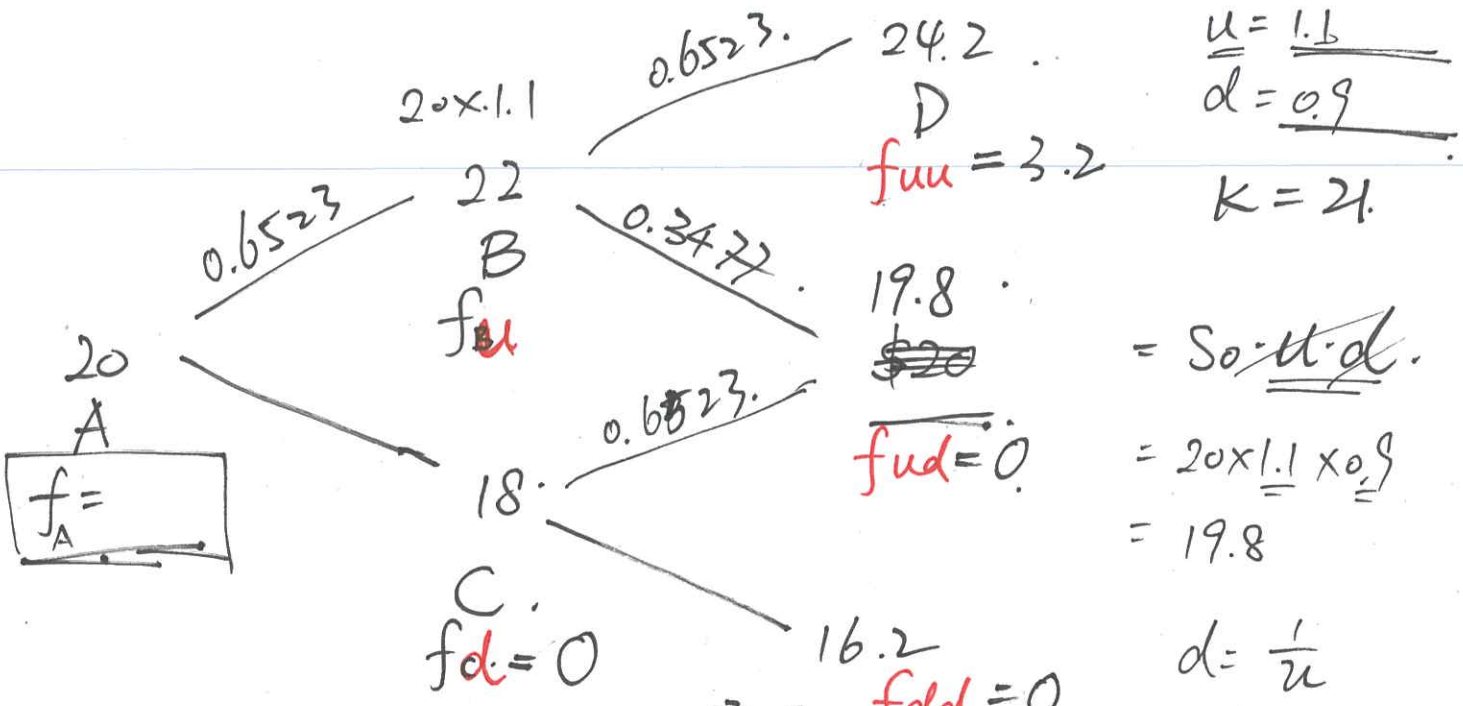
$\pi_0 = 13.5 \cdot e^{-rT} = 12.84$.

$\pi_0 = 10 \times 1 + f = 12.84$ $f = 2.84$.



Exp. Payoff = $6.5 \times 70\% + 0.4 \times 30\%$

(Y)..



$u = 1.1$

$d = 0.9$

$K = 21$

$= S_0 \cdot u \cdot d$

$= 20 \times 1.1 \times 0.9$

$= 19.8$

$d = \frac{1}{u}$

$u \cdot d = 1$

$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{e^{0.12 \times \frac{3}{12}} - 0.9}{1.1 - 0.9}$$

$$= 0.6523$$

B: $f_B = f_u = e^{-0.12 \times \frac{3}{12}} \times (0.6523 \times 3.2 + 0.3477 \times 0)$

$$= 2.0257$$

A: $f = e^{-0.12 \times \frac{3}{12}} \times (2.0257 \times 0.6523 + 0) = 1.2823$

$$f = e^{-rT}(p \cdot f_u + (1-p) f_d)$$